SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA0327	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	K.Mounith	**Reg. No.:**	192425090
Section / Batch:	AIML/2024	**Date:**	16-10-2026

Code of Conduct

I K.Mounith(192425090)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Mounith

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

1. Design a concept map that evaluates the significance of the fundamental concepts of mobile computing.
   Establish relationships among mobility, portability, wireless communication, and ubiquitous computing.
   Prioritize the concepts based on their contribution to mobile computing.
   Justify the linking phrases used between the concepts.

2. Construct a concept map that evaluates the effectiveness of the cellular network architecture.
   Illustrate the interactions among Mobile Station, Base Station Subsystem, and Network Switching Subsystem.
   Identify the most critical components influencing communication.
   Justify your evaluation through meaningful concept relationships.

3. Develop a concept map that evaluates the impact of frequency reuse on cellular network performance.

Relate frequency planning, co-channel interference, adjacent channel interference, and network capacity.
Assess the effectiveness of interference reduction techniques.
Support your evaluation using appropriate linking phrases.

4. Design a concept map that evaluates the importance of different cellular access channels.
   Illustrate how paging, broadcast, control, and traffic channels contribute to communication.
   Rank the channels based on their operational significance.
   Justify the relationships established in the concept map.

5. Construct a concept map that evaluates the contribution of Mobile Station, Base Station Subsystem, and Network Switching Subsystem toward seamless communication.
   Illustrate their interdependencies.
   Assess which subsystem has the greatest influence on service continuity.
   Justify your conclusions using concept relationships.

6. Develop a concept map that evaluates the effectiveness of the mobile network protocol stack.
   Illustrate interactions among protocol layers.
   Assess how each layer contributes to reliable mobile communication.
   Support your evaluation with meaningful hierarchical relationships.

7. Design a concept map that evaluates the role of the core network in delivering mobile communication services.
   Relate authentication, mobility management, subscriber databases, and call routing.
   Determine the most influential core network functions.
   Justify your evaluation through concept linkages.

8. Construct a concept map that evaluates the operational efficiency of a Public Land Mobile Network (PLMN).
   Relate operators, subscribers, switching centers, and radio access networks.
   Assess how these components collectively provide seamless connectivity.
   Justify the relationships included in the concept map.

9. Develop a concept map that evaluates different mobility management functions.
   Illustrate relationships among location update, registration, paging, and tracking.
   Determine which functions most effectively maintain user connectivity.
   Support your evaluation with appropriate concept links.

10. Design a concept map that evaluates various handoff mechanisms used in cellular networks.
    Compare hard handoff, soft handoff, and handoff decision parameters.
    Assess their suitability for maintaining uninterrupted communication.
    Justify your evaluation using interconnected concepts.

11. Construct a concept map that evaluates the effectiveness of roaming mechanisms in mobile communication.
    Illustrate relationships among home network, visited network, authentication, registration, and billing.
    Assess the importance of each process in successful roaming.
    Support your conclusions through logical concept organization.

12. Develop a concept map that evaluates the relationship between mobility, handoff, and roaming.
    Illustrate their dependencies and interactions.
    Determine which mechanism contributes most to uninterrupted communication.
    Justify your evaluation using appropriate linking phrases.

13. Design a concept map that evaluates the influence of cellular performance criteria.
    Relate coverage, capacity, Quality of Service (QoS), reliability, throughput, and delay.
    Assess which criteria most significantly affect user experience.
    Justify the hierarchy established in the concept map.

14. Construct a concept map that evaluates the effectiveness of various cellular system operations.
    Include call setup, authentication, paging, resource allocation, handoff, and call termination.
    Assess the contribution of each operation to overall network performance.
    Support your evaluation through logical concept relationships.

15. Develop a concept map that evaluates the complete mobile communication process.
    Illustrate the interaction among user equipment, access network, core network, mobility management, and service delivery.
    Determine the most critical stages in ensuring communication quality.
    Justify your evaluation with meaningful linking phrases.

16. Design a concept map that evaluates the relationship between frequency reuse, interference, and network capacity.
    Assess how each factor influences cellular system efficiency.
    Prioritize the concepts based on their impact.
    Justify the relationships shown in the concept map.

17. Construct a concept map that evaluates the interdependence among cellular network architecture, protocol stack, and mobility management.
    Assess how these components collectively improve communication reliability.
    Identify the strongest conceptual relationships.
    Justify your evaluation using appropriate linkages.

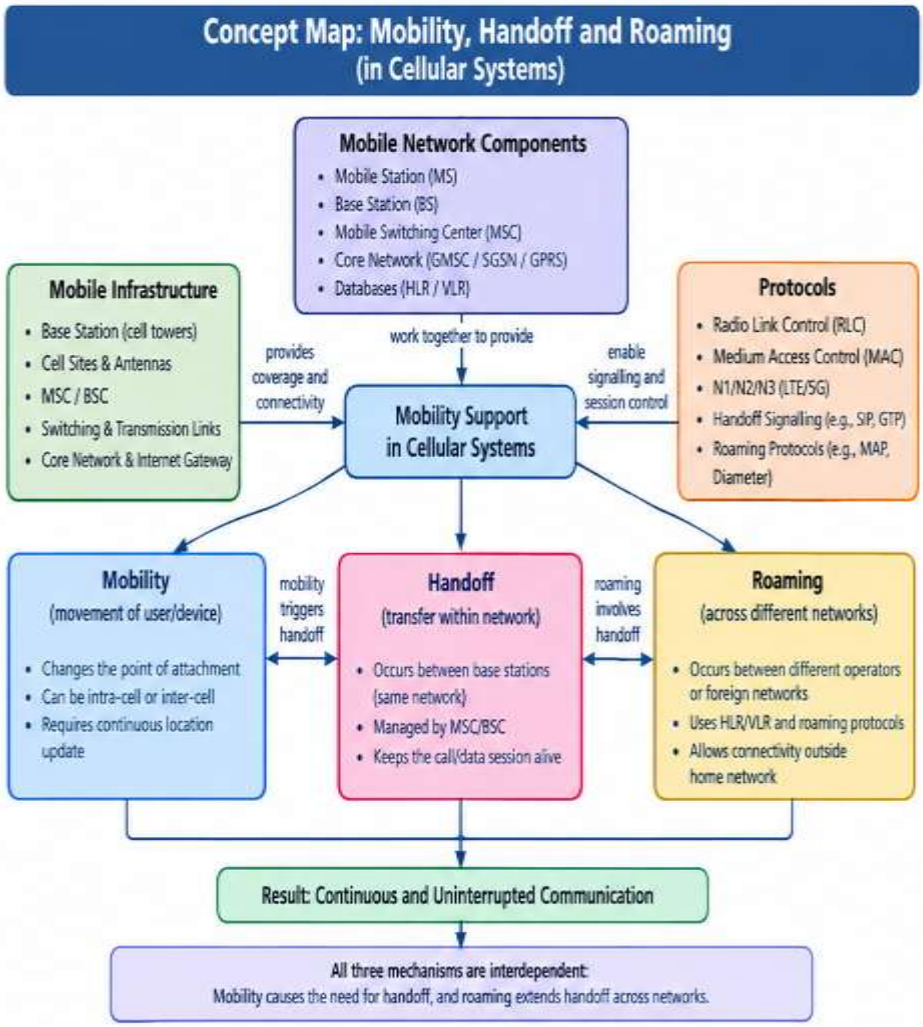
18. Develop a concept map that evaluates resource management strategies in cellular networks.
    Relate channel allocation, power control, interference management, frequency reuse, and handoff management.
    Assess which strategy contributes most to efficient network utilization.
    Support your evaluation through concept relationships.

19. Design a concept map that evaluates the integration of all major concepts in mobile computing.
    Include architecture, protocol stack, mobility management, roaming, handoff, performance criteria, and cellular operations.
    Assess how these concepts collectively support an efficient mobile communication system.
    Justify the hierarchy and linking phrases used.

20. Construct a concept map that evaluates alternative approaches for improving cellular network performance.
    Relate mobility management, handoff optimization, interference reduction, protocol efficiency, and resource allocation.
    Recommend the most effective combination of techniques.
    Justify your recommendation through well-defined conceptual relationships.

ANSWER

12.



Explanation of Relationships, Dependencies and Interactions

1. Mobility → Handoff → Roaming

- **Mobility** is the movement of a mobile user/device from one location to another.
- This movement triggers a **handoff** when the user moves from one cell to another within the same network.
- If the user moves to a different network/operator, **roaming** is required.

2. Dependency

- **Handoff** depends on **mobility** (it happens because of movement).
- **Roaming** depends on both **mobility and handoff** (it is a higher-level mechanism for cross-network movement).
- All three rely on the **mobile infrastructure** (base stations, MSC, core network) and **protocols** (signalling, authentication, location update).

3. Interactions

- **Mobility** provides the trigger (user moves).
- **Handoff** keeps the connection alive within the same network.
- **Roaming** extends the service to other networks using HLR/VLR and roaming protocols.

4. Which Mechanism Contributes Most to Uninterrupted Communication?

Handoff contributes the most to uninterrupted communication.

Justification:

- Handoff is performed quickly (with minimal delay) between base stations.
- It maintains the ongoing call/data session without disconnecting.
- Mobility is the reason for change, and roaming is used only when the user goes to a different network, which may introduce additional delays.
- Therefore, handoff is the key mechanism that ensures seamless and continuous communication within the same network.

According to the Syllabus

Relevant Topics Covered:

- Mobile network components (MS, BS, MSC, HLR, VLR, GMSC, SGSN, GPRS)
- Mobile infrastructures (base stations, cell sites, switching, core network)
- Protocols (RLC, MAC, N1/N2/N3, signalling, roaming protocols)
- Mobility, Handoff and Roaming in cellular systems

Evaluation Rubric (Marks Distribution: 100)

Criteria	Max Marks	Excellent (Full Marks)	Good (Minor Gaps)	Basic (Some Gaps)	Poor (Incorrect)
Understanding of Problem Context	20	Clearly understands all requirements	Minor gaps	Some unclear requirements	Misinterpretation
Application of Concepts	30	Accurately applies concepts	Minor errors	Basic application	Fails to apply
Problem-Solving Approach	20	Logical and efficient	Minor gaps	Some inconsistencies	Illogical/unclear
Accuracy of Solution	20	Completely correct	Minor errors	Partially correct	Incorrect/incomplete
Clarity	10	Well-organized and clear	Minor clarity issues	Limited clarity	Poorly presented

Aim

To analyse the relationship between mobility, handoff and roaming in cellular systems and identify the mechanism that contributes most to uninterrupted communication.

Theory

Mobile computing allows users to access voice and data services while moving from one location to another. It uses mobile stations, base stations, mobile switching centres, network infrastructure and communication protocols to provide connectivity.

1. Mobility: Mobility refers to the movement of a mobile device from one location to another within a cellular network. It allows users to remain connected while travelling.

2. Handoff: Handoff is the process of transferring an ongoing call or data connection from one base station to another. It helps maintain continuous communication when the signal becomes weak or the user moves between cell areas.

3. Roaming: Roaming allows users to access mobile network services through another network or operator when they move outside their home network's coverage area. It requires network agreements and authentication.

Relationship Between Mobility, Handoff and Roaming

- Mobility triggers the need for handoff when a user moves between cell areas.
- Handoff maintains an active connection between base stations.
- Roaming provides network access outside the home network.
- All three mechanisms depend on mobile infrastructure and network protocols.

Result

Handoff contributes most directly to uninterrupted communication because it transfers an active connection between base stations with minimal interruption. Mobility creates the need for handoff, while roaming extends connectivity across different networks. Together, these mechanisms support seamless mobile communication.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach

		approach			
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____